

Simulation study: mixture of experts model to check identifiability

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 2$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.8, 0.2)$
- Degrees of freedom $\boldsymbol{\nu} = (3, 12)$
- Scale matrices of the Wishart distribution Σ_1, Σ_2
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.8, 0.2),
  nus = c(3, 12),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 2

# run a mewishart model
set.seed(123)
fit <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.9, 0.1),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-1562.8 | 618.8 iter/sec
Iter  500 | LL=-1513.1 | 939.5 iter/sec
Iter 1000 | LL=-1514.6 | 958.6 iter/sec
Iter 1500 | LL=-1517.0 | 960.1 iter/sec
Iter 2000 | LL=-1517.6 | 959.7 iter/sec
Iter 2500 | LL=-1519.7 | 971.4 iter/sec
```

```

Iter 3000 | LL=-1514.0 | 977.3 iter/sec
Iter 3500 | LL=-1512.5 | 979.0 iter/sec
Iter 4000 | LL=-1512.4 | 985.9 iter/sec
Iter 4500 | LL=-1512.1 | 990.2 iter/sec
Iter 5000 | LL=-1512.9 | 995.4 iter/sec
Iter 5500 | LL=-1519.9 | 1000.2 iter/sec
Iter 6000 | LL=-1514.5 | 1003.9 iter/sec
Iter 6500 | LL=-1513.2 | 1007.0 iter/sec
Iter 7000 | LL=-1515.3 | 1008.6 iter/sec
Iter 7500 | LL=-1513.0 | 1011.4 iter/sec
Iter 8000 | LL=-1514.9 | 1010.8 iter/sec
Iter 8500 | LL=-1515.8 | 1012.0 iter/sec
Iter 9000 | LL=-1512.6 | 1014.8 iter/sec
Iter 9500 | LL=-1513.5 | 1015.7 iter/sec
Iter 10000 | LL=-1512.5 | 1016.2 iter/sec

```

```
# posterior summaries
```

```
colMeans(fit$pi)
```

```
[1] 0.7906965 0.2093035
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```
apply(fit$nu, 2, mean)
```

```
[1] 3.00674 10.50297
```

```
dat$nu # true nu
```

```
[1] 3 12
```

Check identifiability issue

```
# run the model again with swapped initial weights
```

```
set.seed(123)
```

```
fit2 <- mewishart(S_list,
```

```
  K = my_K_intial, init_pi = c(0.1, 0.9),
```

```
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
```

```
)
```

```

Iter    1 | LL=-1736.3 | 1038.5 iter/sec
Iter   500 | LL=-1513.1 | 1056.1 iter/sec
Iter  1000 | LL=-1514.6 | 1062.0 iter/sec
Iter  1500 | LL=-1517.6 | 1056.4 iter/sec
Iter  2000 | LL=-1517.0 | 1047.2 iter/sec
Iter  2500 | LL=-1519.7 | 1051.3 iter/sec
Iter  3000 | LL=-1513.9 | 1055.3 iter/sec
Iter  3500 | LL=-1512.4 | 1056.4 iter/sec
Iter  4000 | LL=-1512.9 | 1052.9 iter/sec
Iter  4500 | LL=-1512.4 | 1051.3 iter/sec
Iter  5000 | LL=-1511.7 | 1051.7 iter/sec
Iter  5500 | LL=-1515.4 | 1051.6 iter/sec
Iter  6000 | LL=-1515.8 | 1047.5 iter/sec
Iter  6500 | LL=-1512.9 | 1046.5 iter/sec
Iter  7000 | LL=-1516.6 | 1046.2 iter/sec

```

```

Iter 7500 | LL=-1512.9 | 1048.1 iter/sec
Iter 8000 | LL=-1514.9 | 1048.9 iter/sec
Iter 8500 | LL=-1514.3 | 1049.2 iter/sec
Iter 9000 | LL=-1512.3 | 1047.9 iter/sec
Iter 9500 | LL=-1513.6 | 1046.5 iter/sec
Iter 10000 | LL=-1512.5 | 1042.8 iter/sec

```

```
# posterior summaries
```

```
colMeans(fit2$pi)
```

```
[1] 0.7901943 0.2098057
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```
apply(fit2$nu, 2, mean)
```

```
[1] 3.004606 10.359462
```

```
dat$nu # true nu
```

```
[1] 3 12
```

Conclusion:

The two model fittings above with initial weights `init_pi=c(0.9, 0.1)` and `init_pi=c(0.1, 0.9)` result in similar results, which indicate possibly identifiability.

Based on the proof of Proposition 2 in Yakowitz and Spragins (1968), we can probably give similar proofs to show identifiability of the MoE of (inverse) Wishart distributions.